

Lecture_1_743

Group ID: 743

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1 Hand exercises

House	True Price (t)	Predicted Price (y)
1	250	245
2	310	330
3	180	195
4	420	400
5	360	355

1.1 a

Compute MSE of the model on the test set

$$\begin{aligned}MSE &= \frac{1}{5} * ((250 - 245)^2 + (310 - 330)^2 + (180 - 195)^2 + (420 - 400)^2 + (360 - 355)^2) \\&= \frac{1}{5} * (5^2 + (-20)^2 + (-15)^2 + 20^2 + 5^2) \\&= \frac{1}{5} * (25 + 400 + 225 + 400 + 25) \\&= \frac{1}{5} * 1075 \\&= 215\end{aligned}$$

1.2 b

Which house(s) contribute the most to the MSE? Explain why.

House 2 and 4 contribute the most since the difference between the true price and the predicted price is the largest (Difference of 20).

1.3 c

Suppose the prediction for house 4 is 380 instead of 400. By what factor does its contribution to the MSE increase?

$$420 - 400 = 20$$

$$420 - 380 = 40$$

$$20^2 = 400$$

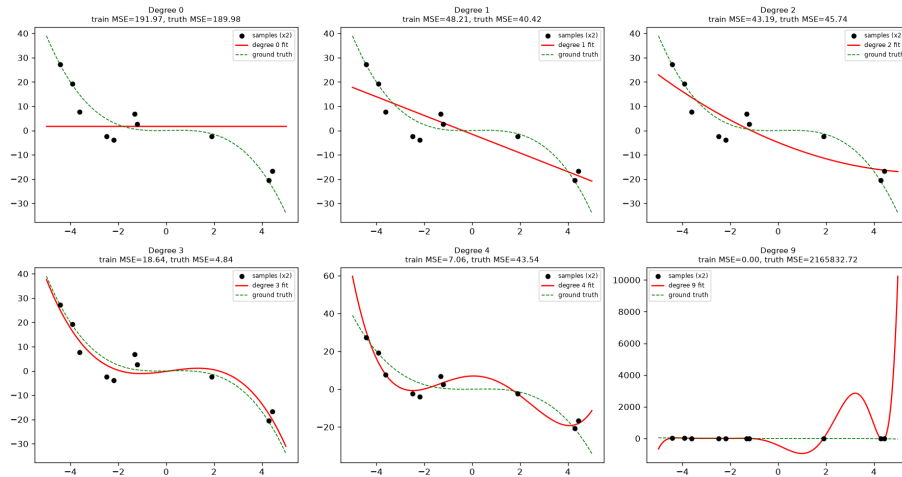
$$40^2 = 1600$$

With a factor of 2 the increase in MSE is 4

2 Other exercises

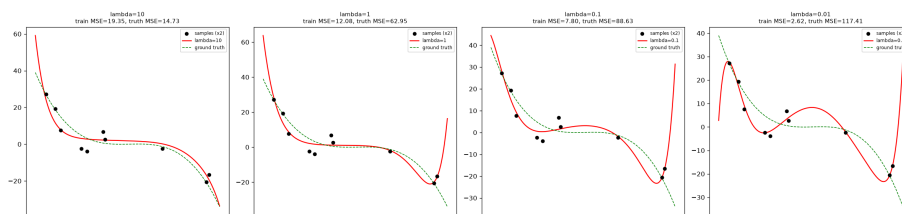
2.1 Exercise 1

As the degree increases the regression gets better until it passes 3 degrees where as it starts to over fit



2.2 Exercise 2

A high lambda value will allow a the regression of a degree 9 to follow the actual regression better



2.3 Exercise 3

With enough sample (in this case 100) even a regression with a degree nine can approximate a regression with degree 3

